# PSX Model & Strategy Performance Report

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Cohort separation: the Task 109 general forward cohort's 43 matured outcomes are NOT LLM performance; the LLM cohort has 0 matured outcomes and probability calibration remains forbidden until >=100 genuine matured LLM outcomes exist.

## Summary comparison

| Model | Status | Promotion |
|---|---|---|
| PatchTST | Rejected/unavailable (resource) | no |
| N-BEATS | Rejected/unavailable (resource) | no |
| TimesNet | Rejected/unavailable (resource) | no |
| Informer | Rejected/unavailable (resource) | no |
| Chronos (Amazon) | Rejected/unavailable (dependency) | no |
| Lag-Llama | Rejected/unavailable (dependency) | no |
| Moirai (Salesforce) | Rejected/unavailable (dependency) | no |
| Time-LLM | Rejected/unavailable (dependency) | no |
| XGBoost | Trained artifact only — no reliable OOS edge evidence | no |
| LightGBM | Trained artifact only — no reliable OOS edge evidence | no |
| CatBoost | Trained artifact only — no reliable OOS edge evidence | no |
| Random Forest | Trained artifact only — no reliable OOS edge evidence | no |
| LSTM | Rejected/unavailable (resource) | no |
| CNN-LSTM | Rejected/unavailable (resource) | no |
| Transformer-LSTM | Rejected/unavailable (resource) | no |
| TabNet | Rejected/unavailable (resource) | no |
| FinGPT | Rejected/unavailable (resource) | no |
| ARIMA | Rejected/unavailable (data) | no |
| Holt-Winters | Rejected/unavailable (data) | no |
| GARCH | Rejected/unavailable (data) | no |
| Ridge Regression | Trained artifact only — no reliable OOS edge evidence | no |
| Kalman Filter | Rejected/unavailable (data) | no |
| Prophet | Rejected/unavailable (dependency) | no |
| GRU | Rejected/unavailable (resource) | no |
| XGBoost (CL) | Duplicate alias — counted once | no |
| LightGBM (CL) | Duplicate alias — counted once | no |
| Random Forest (CL) | Duplicate alias — counted once | no |
| LSTM (CL) | Rejected/unavailable (resource) | no |
| LightGBM-A (CL-2) | Rejected/unavailable (data) | no |

| | | |
|---|---|---|
| LightGBM-B (CL-2) | Rejected/unavailable (data) | no |
| HistGBM (CL-2) | Rejected/unavailable (data) | no |
| ExtraTrees (CL-2) | Rejected/unavailable (data) | no |
| RandomForest (CL-2) | Rejected/unavailable (data) | no |
| Logistic (CL-2) | Rejected/unavailable (data) | no |
| MLP (CL-2) | Rejected/unavailable (data) | no |
| Linear Regression (PI) | Rejected/unavailable (data) | no |
| Logistic Regression (PI) | Rejected/unavailable (data) | no |
| Pattern Matcher KNN (PI) | Rejected/unavailable (data) | no |
| Neural Network (PI) | Rejected/unavailable (data) | no |
| Technical Engine (GM) | Rule strategy — not ML | no |
| Announcements Engine (GM) | Rule strategy — not ML | no |
| Fundamental Engine (GM) | Rule strategy — not ML | no |
| Macro Engine (GM) | Rule strategy — not ML | no |
| LightGBM (CG) | Rejected/unavailable (data) | no |
| XGBoost (CG) | Rejected/unavailable (data) | no |
| Trend Multi-TF (X) | Rule strategy — not ML | no |
| Breakout RelVol (X) | Rule strategy — not ML | no |
| Trend Pullback (X) | Rule strategy — not ML | no |
| RSI Divergence (X) | Rule strategy — not ML | no |
| BB Mean Reversion (X) | Rule strategy — not ML | no |
| Vol Contraction (X) | Rule strategy — not ML | no |
| Relative Strength (X) | Rule strategy — not ML | no |
| Market Regime (X) | Rule strategy — not ML | no |
| Meta-Label CG (X) | Rejected/unavailable (data) | no |
| Triple Barrier Dyn (X) | Rule strategy — not ML | no |
| Gian-Ensemble (GIAN) | Rule strategy — not ML | no |
| Gian-Extender (GIAN-EXT) | Rule strategy — not ML | no |
| M-Ensemble (M) | Rule strategy — not ML | no |
| XGBoost | Research only (no demonstrated reliable positive net edge) | no |
| LightGBM | Research only (no demonstrated reliable positive net edge) | no |
| CatBoost | Research only (no demonstrated reliable positive net edge) | no |
| Random Forest | Research only (no demonstrated reliable positive net edge) | no |
| Ridge Regression | Research only (no demonstrated reliable positive net edge) | no |
| ExtraTrees | Research only (no demonstrated reliable positive net edge) | no |
| HistGradientBoosting | Research only (no demonstrated reliable positive net edge) | no |
| ElasticNet | Research only (no demonstrated reliable positive net edge) | no |
| HuberRegressor | Research only (no demonstrated reliable positive net edge) | no |

| | | |
|---|---|---|
| ExtraTrees | Research only (superseded by Task 100 evaluation) | no |
| HistGradientBoosting | Research only (superseded by Task 100 evaluation) | no |
| ElasticNet | Research only (superseded by Task 100 evaluation) | no |
| HuberRegressor | Research only (superseded by Task 100 evaluation) | no |
| SVR | Research only (superseded by Task 100 evaluation) | no |
| ARIMA | Research only (OOS-evaluated baselines; no net-edge claim) | no |
| Prophet | Research only (OOS-evaluated baselines; no net-edge claim) | no |
| jw_reference_lstm | Research only (reference models; no promotion) | no |
| jw_reference_cnn_lstm | Research only (reference models; no promotion) | no |
| psx_cost_aware_trade_classifier | Shadow only (frozen cost-aware shadow filter) | no |
| psx_net_expectancy_meta_model | Shadow only (frozen cost-aware shadow filter) | no |
| golden_cross_7_30_v1 (Golden Cross 7/30 SMA) | Research only (negative edge after costs) | no |
| task104 portfolio strategies (fixed_first_hit family) | Research only (prospective/shadow) | no |
| volatility-aware overlays (A, B 10/15/20, C) | Research only (predeclared family; 10% target is a hypothesi | no |
| shared strategySignal()/technical-rule implementation + exis | Existing production research signals (unchanged); never auto | no |
| GLM-4.6 multi-source LLM analyst (z-ai-web-dev-sdk) | Shadow only (LLM cohort distinct from the general Task 109 c | no |

## Account-level comparison (PKR 100,000)

| Strategy | End balance | ROI % | Max DD % | Trades |
|---|---|---|---|---|
| golden_cross_baseline | 62208.46 | -37.792 | -87775.26 | 365 |
| overlay_B_vol_target_10pct | 98060.14 | -1.94 | -23.621 | 430 |
| overlay_B_vol_target_15pct | 93817.25 | -6.183 | -33.656 | 430 |
| overlay_B_vol_target_20pct | 88763.21 | -11.237 | -42.54 | 430 |
| overlay_A_fixed_vol_exposure | 81607.67 | -18.392 | -62.414 | 430 |
| overlay_C_expansion_protection | 81607.67 | -18.392 | -62.414 | 430 |
| cash | 100000.0 | 0.0 | 0.0 | 0 |
| buy_hold_equal_weight | 120794.06 | 20.794 | None | None |

Task 114 forward-horizon maturity — frontier 2026-11-04, new sessions since 2026-11-03: ["2026-11-04"]; 0 matured 5/10/21-session outcomes in every cohort (reported honestly). Early SL/TP3 exits are separate measurements, never horizon maturities. LLM calibration stays unavailable until >=100 genuine matured LLM outcomes.

| Cohort | Obs | Mat5/Pend5 | Mat10/Pend10 | Mat21/Pend21 | Term | Unfill | Excl |
|---|---|---|---|---|---|---|---|
| task105-gen1 | 12 | 0/12 | 0/12 | 0/12 | 28 | 0 | 0 |
| task106-gen2 | 22 | 0/22 | 0/22 | 0/22 | 27 | 0 | 0 |
| task107-gen3 | 23 | 0/23 | 0/23 | 0/23 | 25 | 0 | 0 |
| task111-llm-multisource-gen1 | 23 | 0/23 | 0/23 | 0/23 | None | 0 | 48 |

Full profiles, per-horizon metrics, cost assumptions and evidence lineage: see docs/reports/psx_model_strategy_performance_report.md and

data/reports/model_strategy_performance_latest.json. Missing values are marked 'Not available in source evidence'.